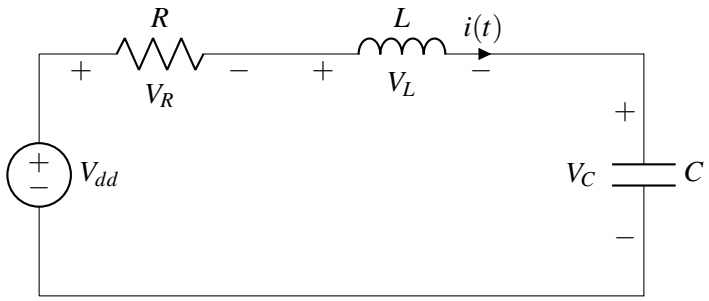
CSM
Spring 2026

EECS 16B
Worksheet 8

1. Introduction to Second Order Differential Equations

Take a RLC circuit in series. We will examine the behavior of this circuit with different R, L, and C values.



(a) Using your knowledge of I-V relationships of inductors and capacitors, write out a KVL equation that represents the behavior of this circuit.

Solution:

$$-V_{dd} + V_L + V_R + V_C = 0$$

$$V_L = L\frac{dI}{dt} = LC\frac{d^2V_C}{dt^2}$$

$$C\frac{dV_C}{dt}$$

$$V_R = RI = RC\frac{dV_C}{dt}$$

$$LC\frac{d^2V_C}{dt^2} + RC\frac{dV_C}{dt} + V_C = V_{dd}$$

(b) What is the steady state solution of $V_C(t)$?

Solution: At steady state, $\frac{dV_C}{dt} = 0$, so we can set $V_{dd} = V_C(\infty)$.

(c) Solve for the solution of the homogenous differential equation.

Solution: We can guess an exponential solution using the characteristic polynomial. We replace $\frac{dV_C}{dt}$ as r:

$$LCr^2 + RCr + 1 = 0$$

Dividing the equation by $\frac{1}{LC}$, we get

$$r^2 + \frac{R}{L}r + \frac{1}{LC} = 0$$

Solving for r,

$$r = \frac{-\frac{R}{L} \pm \sqrt{(\frac{R}{L})^2 - \frac{4}{LC}}}{2}$$

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$$V_C(t) = C_0 e^{\frac{-\frac{R}{L}+\sqrt{(\frac{R}{L})^2-\frac{4}{LC}}}{2}t} + C_1 e^{\frac{-\frac{R}{L}-\sqrt{\frac{R}{L}^2-\frac{4}{LC}}}{2}t}$$

Where C_0 and C_1 are determined by the initial conditions of V_C.

(d) Let's analyze our solution. Under what conditions will we

i. Have two real exponent terms?
Solution:

$$\sqrt{(\frac{R}{L})^2 - \frac{4}{LC}} > 0$$

Under these conditions, our circuit will experience **overdamping**, better written as $\sqrt{\frac{R}{2L}^2 - \frac{1}{LC}} > 0$.

ii. Have only one exponent term?
Solution:

$$\sqrt{(\frac{R}{L})^2 - \frac{4}{LC}} = 0$$

Under these conditions, our circuit will experience **critical damping**.

iii. Have two complex exponent terms?
Solution:

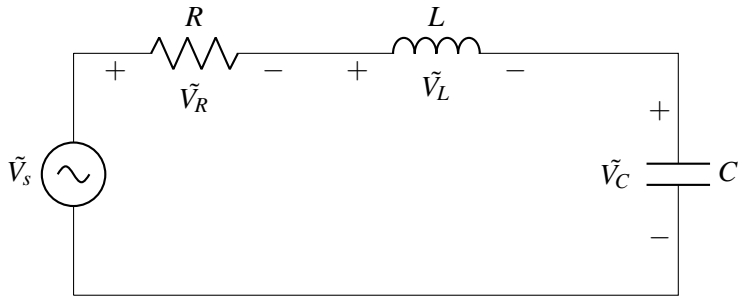
$$\sqrt{(\frac{R}{L})^2 - \frac{4}{LC}} < 0$$

Under these conditions, our circuit will experience **underdamping**.

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2. RLC Circuit

In this question, we will take a look at an electrical systems described by second-order differential equations and analyze it in the phasor domain. Consider the circuit below where $\tilde{V}_s$ is a sinusoidal signal, $L = 1\,\text{mH}$, and $C = 1\,\text{nF}$:



(a) **Transform the circuit into the phasor domain.**

Solution:

$$Z_R = R$$
$$Z_L = j\omega L$$
$$Z_C = \frac{1}{j\omega C}$$

(b) **Solve for the transfer function $H_C(\omega) = \frac{\tilde{V}_C}{\tilde{V}_s}$ in terms of R, L, and C.**

Solution:

$\tilde{V}_C$ is a voltage divider where the output voltage is taken across the capacitor.

$$\tilde{V}_C = \frac{Z_C}{Z_R + Z_L + Z_C}\tilde{V}_s,$$
$$H_C(\omega) = \frac{Z_C}{Z_R + Z_L + Z_C} = \frac{\frac{1}{j\omega C}}{R + j\omega L + \frac{1}{j\omega C}}.$$

Multiplying the numerator and denominator by $j\omega C$ gives

$$H_C(\omega) = \frac{1}{(j\omega)^2 LC + j\omega RC + 1}$$

(c) **Solve for the transfer function $H_L(\omega) = \frac{\tilde{V}_L}{\tilde{V}_s}$ in terms of R, L, and C.**

Solution:

$\tilde{V}_L$ is a voltage divider where the output voltage is taken across the inductor.

$$\tilde{V}_L = \frac{Z_L}{Z_R + Z_L + Z_C}\tilde{V}_s,$$
$$H_L(\omega) = \frac{Z_L}{Z_R + Z_L + Z_C} = \frac{j\omega L}{R + j\omega L + \frac{1}{j\omega C}}.$$

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Multiplying the numerator and denominator by $j\omega C$ gives

$$H_L(\omega) = \frac{(j\omega)^2 LC}{(j\omega)^2 LC + j\omega RC + 1}$$

(d) **Solve for the transfer function $H_R(\omega) = \frac{\widetilde{V}_R}{\widetilde{V}_s}$ in terms of R, L, and C.**

Solution:

$\widetilde{V}_R$ is a voltage divider where the output voltage is taken across the resistor.

$$\widetilde{V}_R = \frac{Z_R}{Z_R + Z_L + Z_C}\widetilde{V}_s,$$
$$H_R(\omega) = \frac{Z_R}{Z_R + Z_L + Z_C} = \frac{R}{R + j\omega L + \frac{1}{j\omega C}}.$$

Multiplying the numerator and denominator by $j\omega C$ gives

$$H_R(\omega) = \frac{j\omega RC}{(j\omega)^2 LC + j\omega RC + 1}.$$

(e) **Sketch the Bode plots of $H_C(\omega)$, $H_L(\omega)$, and $H_R(\omega)$ when $R = 2\,\text{k}\Omega$.**

Solution: $H_C(\omega)$:

Let's look at the poles. It is a second order transfer function of the form $\left(\frac{j\omega}{\omega_c}\right)^2 + j\omega\frac{2\xi}{\omega_c} + 1$, where $\omega_c = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-3}10^{-9}}} = 10^6$. There are two frequency terms in the denominator so there are two poles. This means that there are 2 poles at $\omega_c = 10^6$.

Zeros:

Since the numerator has no frequency terms, there are no zeros.

We have 2 poles and no zeros, which means that this will be a low pass filter with a cutoff frequency of $\omega = 10^6$, with a -20 dB/decade rolloff *per* pole. This means we have a -40 db/decade rolloff at $\omega = 10^6$.

To find the magnitude in the flatband for a low pass filter, we look at the magnitude when all frequency terms are 0. For this filter, we just get a magnitude of $\frac{1}{1} = 1$, or 0 dB.

To find the cutoff frequency, we need to set this into the rational transfer function form so that we can easily solve for $H_C(j\omega) = \frac{1}{\sqrt{2}}$. In.other words, we can model it in the form below to later pattern match what our cutoff frequency is .

$$(\frac{1}{1 + \frac{j\omega}{\omega_c}})^2 = \frac{1}{(\frac{j\omega}{\omega_c})^2 + j\omega\frac{2E}{\omega_c} + 1}$$

where ω_c is our cutoff frequency and E is an expression involving R, C, and L. In this form, we can see that $\omega_c = \sqrt{1/LC} = 10^6$

As for phase, the cut off frequency at $\omega_c = 10^6$ will cause a phase shift starting at $10^5\omega$ (one decade before ω_c) until a decade $10^7\omega$ (one decade after ω_c) past the cutoff frequency. This results in the Bode plot below.

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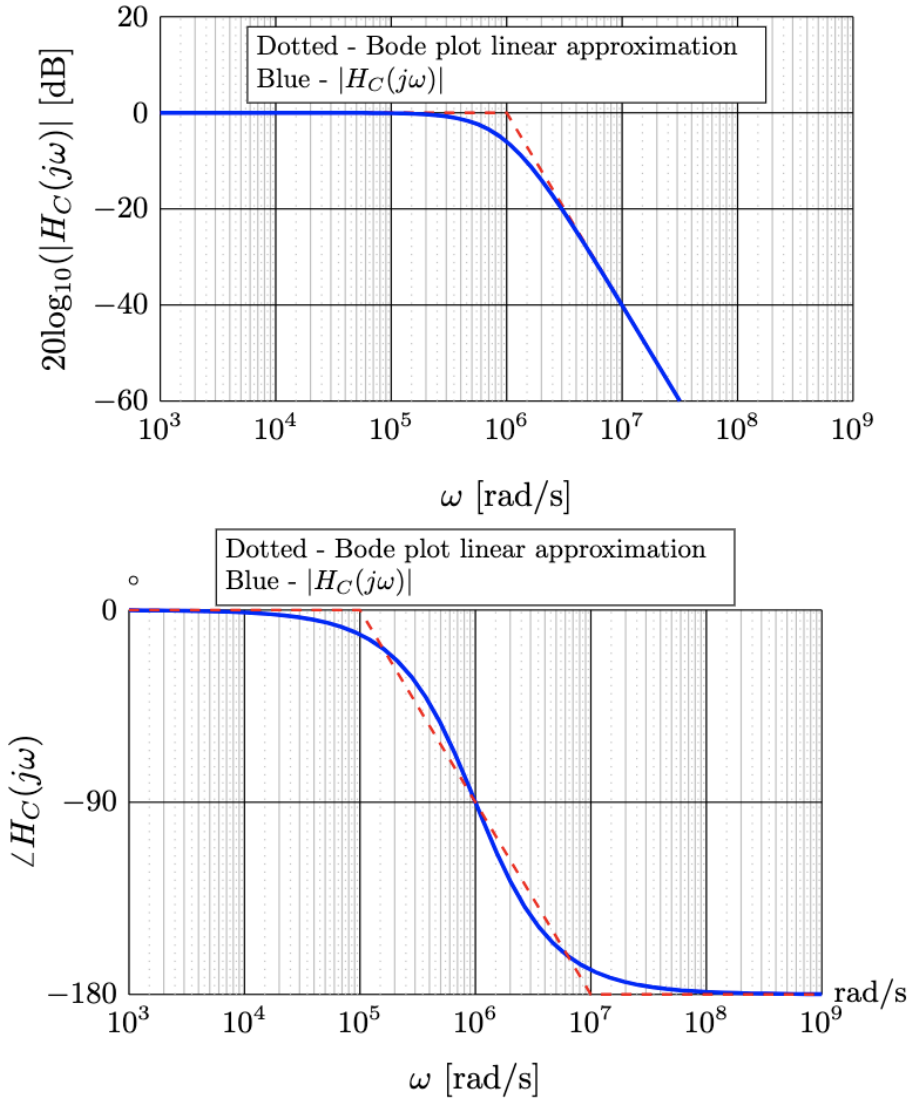


Figure 1: Bode plot of $H_C(\omega)$

$H_L(\omega)$:

Poles: same as $H_C(\omega)$, 2 poles at $10^6 \frac{\text{rad}}{\text{s}}$.

Zeros: Since there is only a frequency term, the zero is located at $0 \frac{\text{rad}}{\text{s}}$. Since ω is squared, there are 2 zeros located at $0 \frac{\text{rad}}{\text{s}}$. This causes an initial phase shift of 180° (90° per zero at $0 \frac{\text{rad}}{\text{s}}$).

This results in a high pass filter with a corner frequency at $10^6 \frac{\text{rad}}{\text{s}}$, where the roll off is $40 \frac{\text{dB}}{\text{dec}}$.

To find the magnitude in the flatband for a high pass filter, we look at the magnitude when everything but the highest order frequency terms is zero. Since this transfer function is second order, we look at the ω^2 terms and get a magnitude of $\frac{(j\omega)^2 LC}{(j\omega)^2 LC} = 1$.

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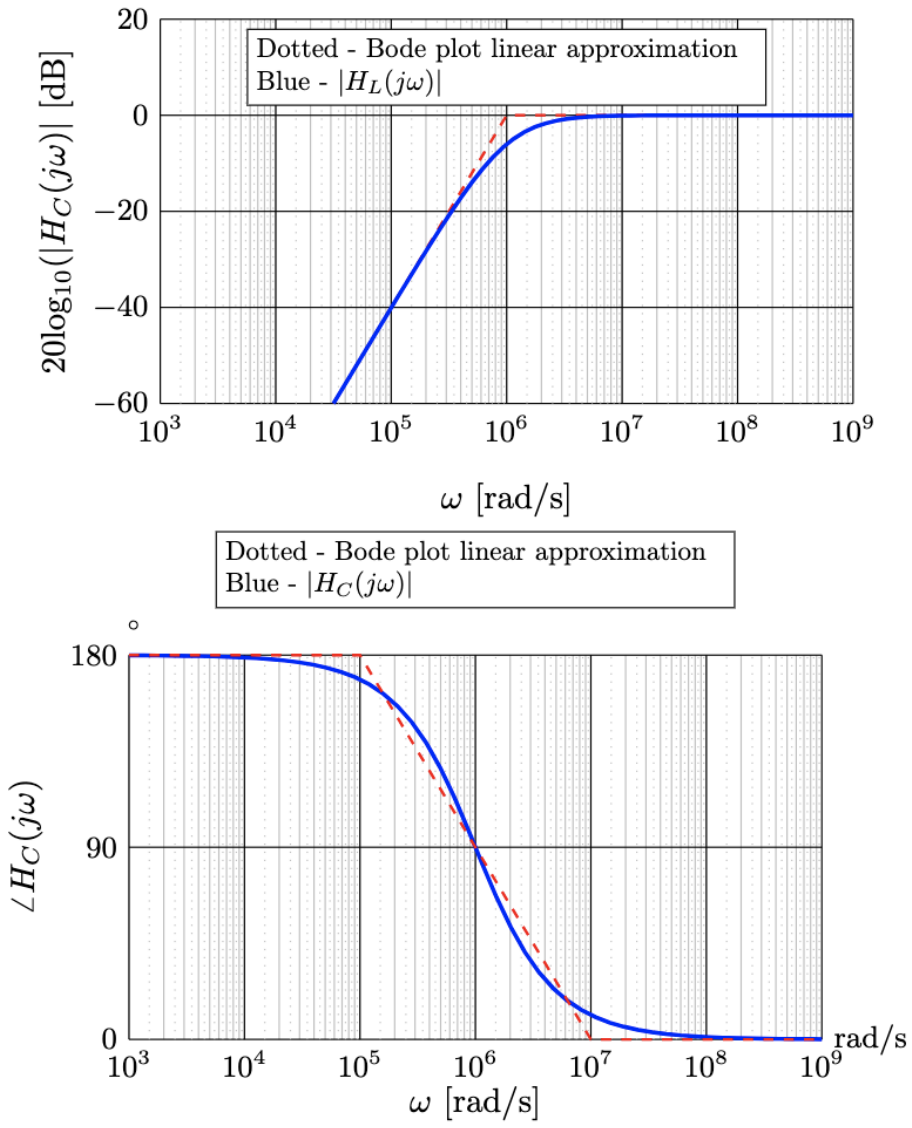


Figure 2: Bode plot of $H_L(\omega)$

$H_R(\omega)$:

Poles: same as others, 2 poles at $10^6 \frac{\text{rad}}{\text{s}}$.

Zeros: Like $H_L(\omega)$, it has a zero at $0 \frac{\text{rad}}{\text{s}}$, but this time there is only 1.

This creates a bandpass filter. To find the peak magnitude, we look at the first asymptote. Before ω_c, the only pole or zero that affects $|H(\omega)|$ is the zero at $0 \frac{\text{rad}}{\text{s}}$:

$$|H_R(\omega \leq \omega_c)| = |j\omega RC|$$

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Plugging in $\omega = \omega_c$ will give us the peak value of the transfer function:

$$|H_R(\omega = \omega_c)| = |j\omega_c RC| = \left|\frac{jRC}{\sqrt{LC}}\right| = \left|\frac{jR\sqrt{C}}{\sqrt{L}}\right|$$

$$|H_R(\omega = \omega_c)| = \frac{2000\,\Omega\sqrt{10^{-9}\text{F}}}{\sqrt{10^{-3}\text{H}}} = 2$$

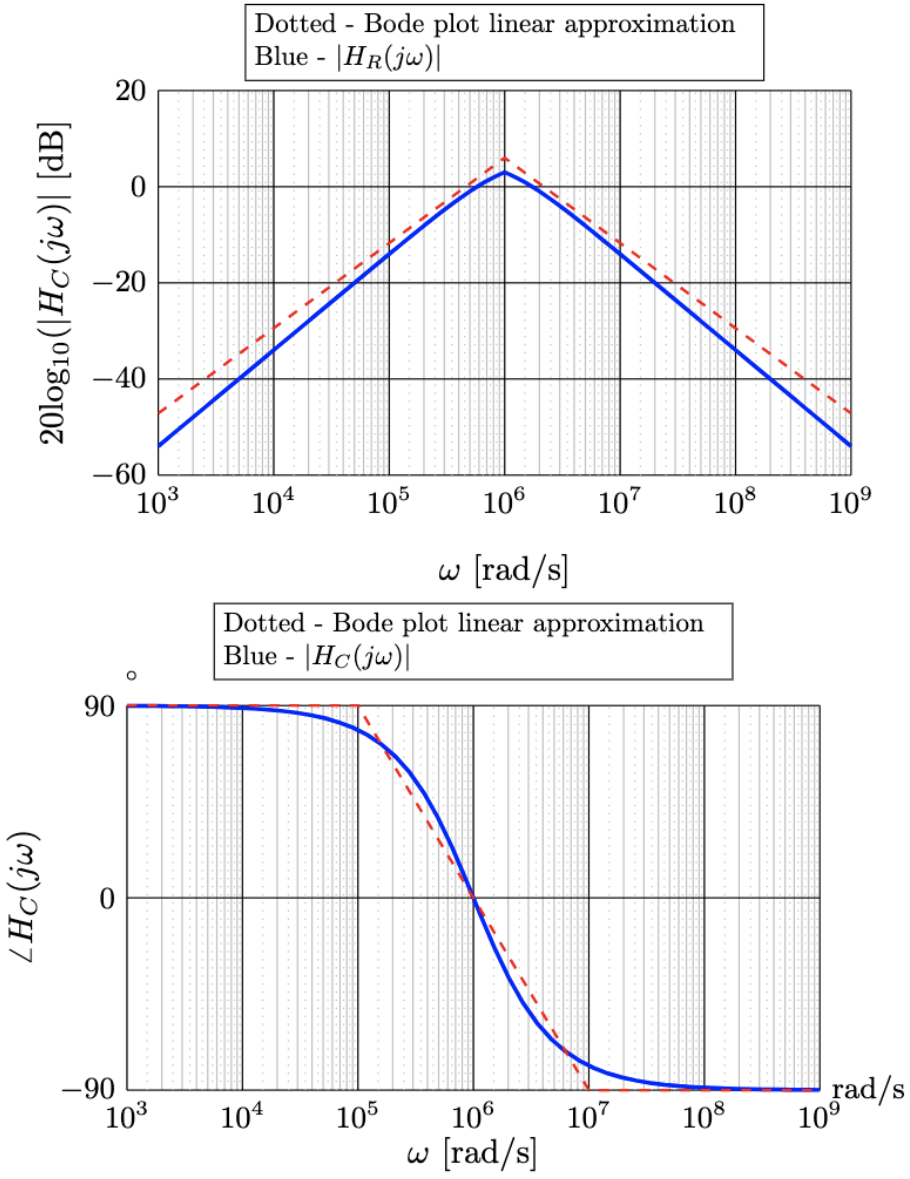


Figure 3: Bode plot of $H_R(\omega)$

(f) **(Practice) Draw the Bode plot of $H_R(\omega)$ two more times, but change R to be $20\,\Omega$, then $200\,\text{k}\Omega$.**
Solution:

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Looking back at our solution in part (e), the only difference is the peak value of $H(\omega)$. Plugging the new resistor values in, we get:

$$R = 20\,\Omega \rightarrow |H(\omega_c)| = \frac{20\sqrt{10^{-9}}}{\sqrt{10^{-3}}};$$
$$R = 200\,\text{k}\Omega \rightarrow |H(\omega_c)| = \frac{2 \cdot 10^5\sqrt{10^{-9}}}{\sqrt{10^{-3}}}.$$

Every other part of the Bode plot is the same as the plot in part (e).

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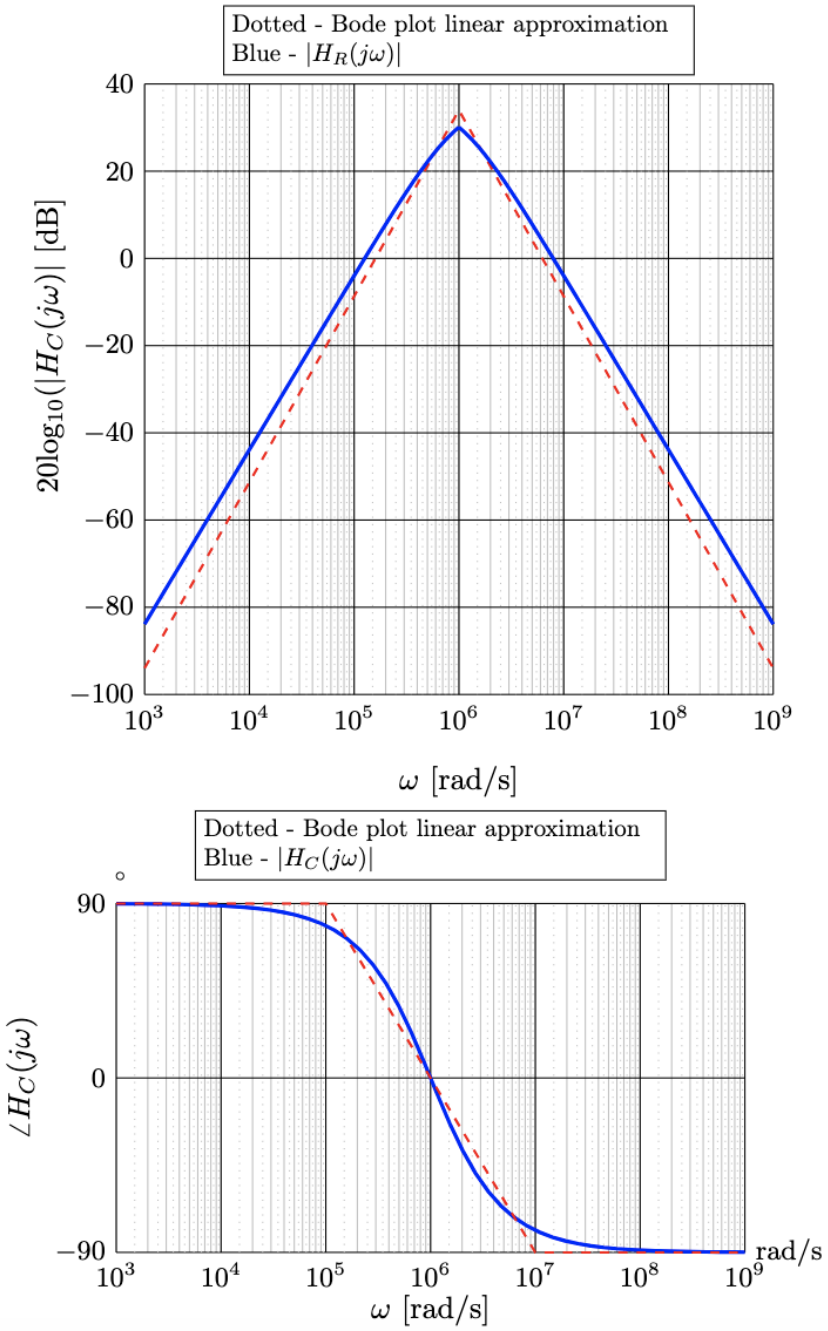


Figure 4: Bode plot of $H_r(\omega)$ for $R = 20\Omega$

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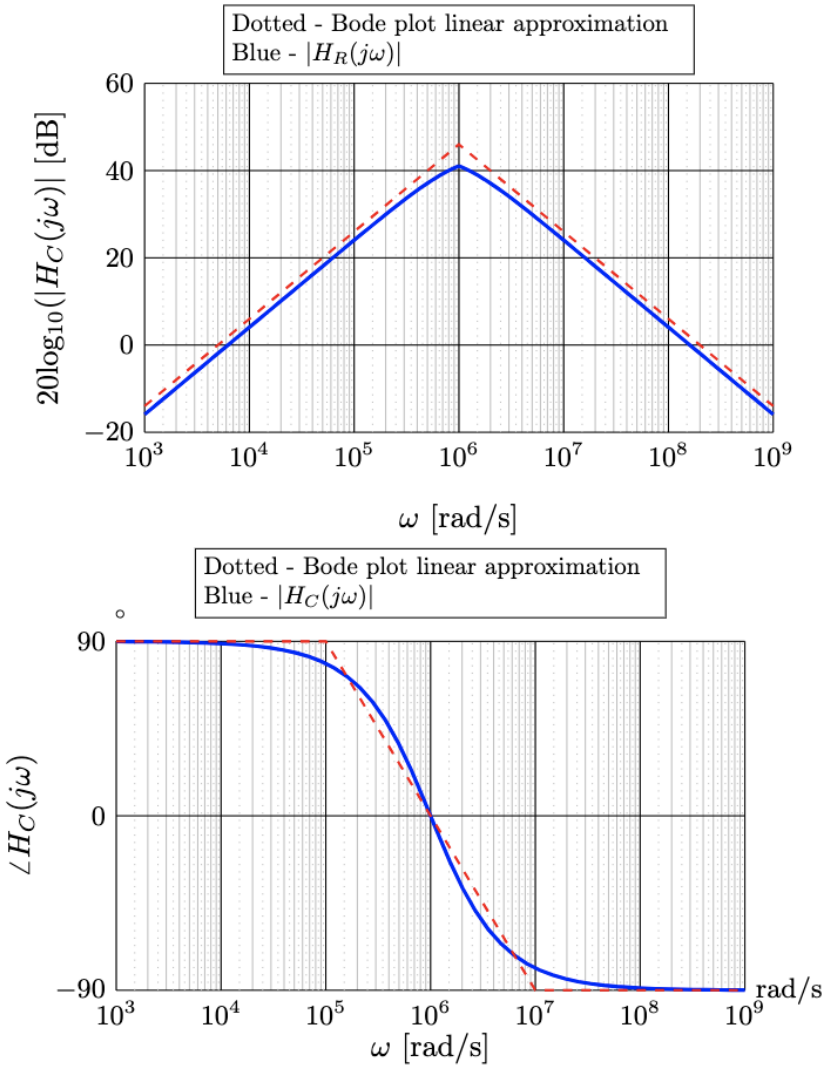
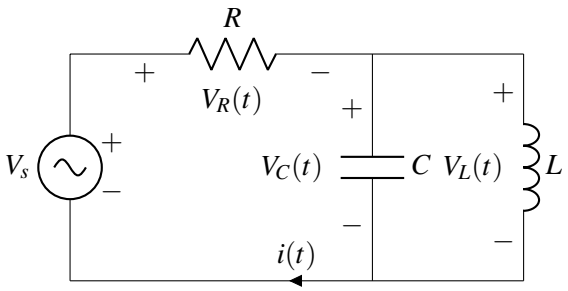


Figure 5: Bode plot of $H_C(\omega)$ for $R = 200\Omega$

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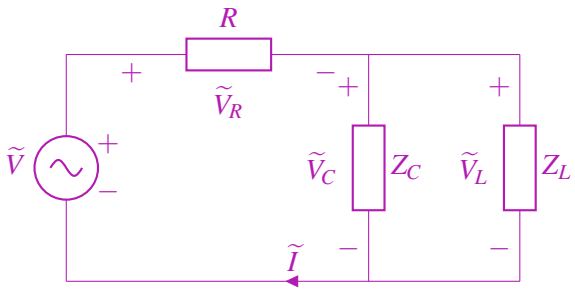
3. Power Analysis

Consider the following circuit:



$\omega = 1000\,\frac{\text{rad}}{\text{s}}$, $V_s(t) = 1000\sqrt{2}\cos\left(1000t + \frac{\pi}{2}\right)$, $R = 1000\,\Omega$, $C = 1\,\mu\text{F}$, $L = 0.5\,\text{H}$.

(a) First we need to transform the whole circuit from time domain to phasor domain. Please **redraw the circuit** with only $\widetilde{V}$, $\widetilde{I}$, and all impedances of the circuit components.
Solution:



(b) **Find the phasor $\widetilde{I}$ through the circuit and the total impedance Z_{tot} of the circuit.** **Solution:** To find the phasor current $\widetilde{I}$, we need to solve for the total impedance of the circuit and use Ohm's Law in phasor form. The impedance of the circuit is given by

$$\begin{aligned} Z_{tot} &= R + \frac{1}{\frac{1}{Z_L} + \frac{1}{Z_C}} \\ &= R + \frac{1}{\frac{1}{j\omega L} + \frac{1}{\frac{1}{j\omega C}}} \\ &= 1000 + \frac{1}{\frac{1}{j(1000)(0.5\text{H})} + \frac{1}{\frac{1}{j(1000)10^{-6}\text{F}}}} \\ &= 1000 + 1000\,j \\ &= 1000\sqrt{2}e^{\frac{\pi}{4}j}. \end{aligned}$$

The phasor voltage can be represented in Euler's form as $\widetilde{V}_s = 1000\sqrt{2}e^{\frac{\pi}{2}j}$.
The phasor current $\widetilde{I}$ can then be found as

$$\begin{aligned} \widetilde{I} &= \frac{\widetilde{V}_s}{Z_{tot}} \\ &= \frac{1000\sqrt{2}e^{\frac{\pi}{2}j}}{1000\sqrt{2}e^{\frac{\pi}{4}j}} \\ &= e^{\frac{\pi}{4}j}. \end{aligned}$$

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(c) **Find the real power, reactive power, and apparent power.** *Hint:*

$$\text{apparent power} = \frac{1}{2}|\widetilde{V}||\widetilde{I}| \tag{1}$$

$$\text{reactive power} = \frac{1}{2}|\widetilde{V}||\widetilde{I}|\sin\big(\angle(Z_{tot})\big) \tag{2}$$

$$\text{real power} = \frac{1}{2}|\widetilde{V}||\widetilde{I}|\cos\big(\angle(Z_{tot})\big) \tag{3}$$

Solution: The real power is

$$P = \frac{1}{2}|\widetilde{V}||\widetilde{I}|\cos\big(\angle(Z_{tot})\big) = 0.5\,\text{kW}$$

The reactive power is

$$Q = \frac{1}{2}|\widetilde{V}||\widetilde{I}|\sin\big(\angle(Z_{tot})\big) = 0.5\,\text{kVAR}$$

The apparent power is $\frac{1}{2}|\widetilde{V}||\widetilde{I}| = 1.414\,\text{kVA}$.

(d) Find the power factor, and state whether it is lagging or leading. *Hint:*

$$\text{power factor} = \frac{\text{real power}}{\text{apparent power}} = \cos(\theta_d),$$

where θ_d is phase difference betweent voltage and the current of the circuit. **Solution:** power factor $=$ cos(phase difference between voltage and current) $= \cos\big(|\angle(Z_{tot})|\big) = \cos\big(\frac{\pi}{4}\big) = \frac{\sqrt{2}}{2}$.
Two essential properties we need to be aware of:
1. If the current reaches its peak value up to $\frac{\pi}{2}$ later than the voltage, the power factor is lagging, and the circuit's load is inductive.
2. If the current reaches its peak value up to $\frac{\pi}{2}$ ahead of the voltage, the power factor is leading, and the circuit's load is capacitive.
From the phasor representation to the time domain, we know that $i(t) = \cos\big(1000t + \frac{\pi}{4}\big)$. The phase of $V_s(t)$ is $\frac{\pi}{2}$, and the phase of $i(t)$ is $\frac{\pi}{4}$. One way to check when the current or voltage reaches its peak value is just to check when the term inside cos becomes 2π. In this case, when $t = \frac{\frac{7}{4}\pi}{1000}$, the current reaches its maximum. However, when $t = \frac{\frac{6}{4}\pi}{1000}$, the voltage reaches its maximum. The current comes later, so the power factor is lagging. Alternatively, based on the value of reactive power, we know that it is positive, so the power factor is lagging.

(e) Based on the previous question, **is the circuit's load resistive, inductive, or capacitive?** **Solution:** Since the power factor is lagging, the cicuit's load is inductive.

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